

Measurements and Information

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Abstract

This technical note exposes the inter-relation between measurements (or, more specifically, readings done by an automated instrument containing a digital computer) and "information".

The question addressed here is: How much of this "information" could be present in a data sample?

Introduction

Let $M = \{(t_i, m_i)\}_{i=1}^n$ be a finite sequence of timed readings taken using an instrument characterized by:

- A finite resolution δm .
- A finite and constant sampling rate $f = \frac{1}{\delta t}$ with $\delta t = t_{i+1} - t_i$ ($i = 1, 2, \dots, n-1$).
- A finite output set $S \subset \mathbb{R}$ such that $m_i \in S$ for $i = 1, 2, \dots, n$.

Note: I've written (t_i, m_i) to signify the i -th "reading" assuming t_i represents the "time stamp" referred to a specific instant (with time instants t_i forming a monotonically increasing sequence of real numbers), and m_i the corresponding instrument reading.

Now, let $\Delta t := t_n - t_1$ be the "averaging time". Assuming all individual readings are "valid" (that is, good for being considered with the same weight of importance), we can define the measurement

$$m = \frac{1}{n} \sum_{i=1}^n m_i \tag{1}$$

My question is, how much information is potentially contained in m , depending on δm , δt and Δt .

I urge you to consider this definition does *not* refer to whether the individual readings concurring to it are correct or not. Accuracy has not been included, and this is intentional.

Last: I gave no references. All I'm about to use is well known in literature since decades, and is now part of the engineers' and physicists' commonsense. Cluttering the text with academic references all older than 1970 would have just cluttered the text of what, after all, is not a research paper - nor it claims any originality.

What "information" means, in this lab note

As a measurement of the quantity of information I am using Shannon's *entropy*,

$$E = - \sum_{\alpha} p_{\alpha} \log p_{\alpha} \quad (2)$$

with the summation extended to all possible system states α , and p_{α} their probability, or empirical frequency (here and in the following sections the logarithm is intended in base 2).

Explicit expression

A well-known result from information theory states the entropy E attains its maximum when all states are equally probable.

In the context of this lab note, a "state" is one specific sequence M , any two same-length sequences M' , M'' being distinct as long as there exist one $i = 1, 2, \dots, n$ with $m'_i \neq m''_i$. Assuming this definition, all "states" ("distinct sequences") are mutually exclusive, so

$$\sum_{\alpha} p_{\alpha} = 1 \quad (3)$$

that is,

$$p_{\alpha} = \frac{1}{N} \quad (4)$$

where N denotes the total number of states. This known, we can say

$$\hat{E} = -N \cdot \frac{1}{N} \log \frac{1}{N} = \log N \quad (5)$$

And this is the gross potential information one can expect to find in a data sequence under the assumptions stated in the introduction.

Now, all what remains to do is evaluating N .

Let $q \in \mathbb{N}$ be the number of possible values the reading m_i can take. Then,

$$N = q^n \quad (6)$$

and

$$\hat{E} = n \cdot \log q \quad (7)$$

The specific value of q can be assumed, if the set S has a complicate shape. But if S is an interval (by far the most common case), then we can say

$$q \cong \frac{m(S)}{\delta m} \quad (8)$$

where I indicated with $m(S)$ the *measure* of S , that is $\sup S - \inf S \in \mathbb{R}$.

From an electronic engineer's perspective, the value of q can also be obtained from the number of bits used to encode any reading m_i . Said this b : then

$$q = 2^b \quad (9)$$

and

$$\hat{E} = n \cdot b \quad (10)$$

This makes a lot of sense: it is the number of bits used when encoding one entire sequence.

In the practice of instrumentation and measurements, however, it may happen the full measurement range is strictly contained within the overall range covered by the full extension of the interval conceded by the nominal bit length and sensor resolution. In this case, using the estimate (8) may be more effective.

The information destruction effect

The final measurement value, m , is expressed as a real number. This is fine, if we limit ourselves to mathematical considerations.

But in current practice, the value of m is stored in the memory of a computer, or transmitted over a finite capacity communication line, or both.

Add to this aesthetically considerations, like "if the readings have two significant digits then their mean should also be expressed in two figures".

In mathematical terms, this can be expressed by saying the value of m is encoded using d bits of information, usually subject to a condition $d \ll n \cdot b$.

The difference

$$\Delta_i := n \cdot b - d \tag{11}$$

represents the information available which is not considered, or, if you prefer, is lost.

The fact reduction operations like a mean always involve some destruction of information is well known among computer scientists, and accepted as an unavoidable evil.

In the context of environmental measurement engineering, however, we should not accept blindly this fatalism: the value of a measurement could be potentially very high, literally life-saving.

My suggestion, of pragmatic nature, is to always encode measurements at the top bit length available: all bits not included cannot be recovered, and are lost forever.

If you then have some aesthetic scruple, you may give voice to it when publishing data in, say, a report table or a web site. The manufacturers of data loggers aimed at scientific applications always use the maximum bit length allowed by their computing cores. This is not by chance.

An example

To conclude this lab note I will exhibit an example, taken from my professional world.

Let's imagine the sequences are hourly measurements of wind speed, obtained from an engineering-grade 3D ultrasonic anemometer (nominal data rate 10 Hz, $q = 6000$ possible values).

In this case, any sequence contains 36000 wind speed values, each taking

$$\log_2 q = \log_2 6000 = 12.551 \quad (12)$$

bits.

The information potentially present in the sequence is then

$$36000 \cdot 12.551 = 451826.9 \quad (13)$$

bits.

Now consider the same measurement, this time taken using a cup anemometer sampled every 10s (this is a typical value). For comb-type cup anemometers the resolution depends on wind speed, but, let's assume it always is maximum, with a speed range between 0 and 60 m/s, as the sonic anemometer's.

This time, the information stored in the sequence is

$$360 \cdot 12.551 = 4518.269 \quad (14)$$

This is 100 times less than the ultrasonic anemometer's. Regardless, as I said in the Introduction, of the measurement accuracy and any other consideration on cup anemometers (starting speed, nonlinear zone, ...).

Does this elicit practical considerations? Of course, yes - in my own view:

- Once is decided how many figures you retain when storing wind speed data into a database, the information destruction you fatally incur in is much larger in case of the ultrasonic anemometer.
- Cup-vane wind speed may be safely presented using one decimal digit. For ultrasonic, I recommend *at least three*.
- As the width of confidence limits is related to the total variation in data divided by the square root of the sample size,